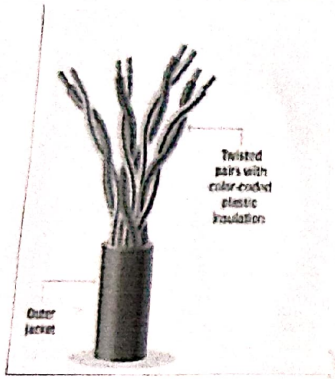
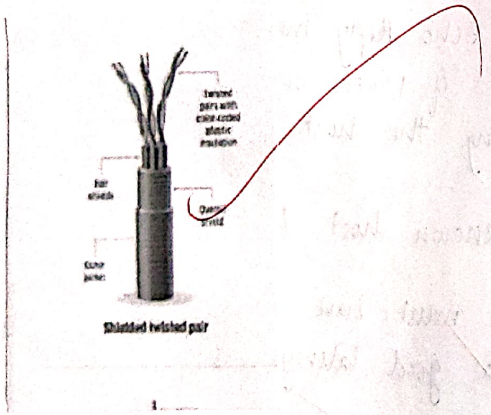




16/7/26

EX 2 : STUDY OF DIFFERENT TYPES OF NETWORK CABLES

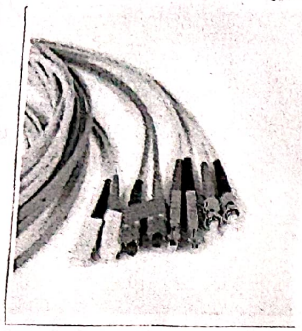
AIM: To study and understand different types of network cable, To make your own ethernet cross-over cable / straight cable

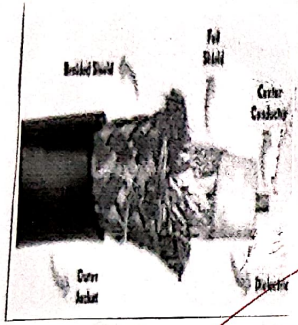
a. Understand different types of Network Cable

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable.

1. UTP

Category	Max. data Transmissin	Advantage / Disadvantage	Application / Use
Category 3	10 bps	Advantages. × cheaper in cost. × easy to install as they have a smaller overall diameter. Disadvantage · v More prone to EMI and noise	10 Base -T Ethernet
Category 5	upto 100bps		Fast ethernet, Gigabit Ethernet
Category 5e	1 Gbps		Fast ethernet, Gigabit ethernet





2 STP

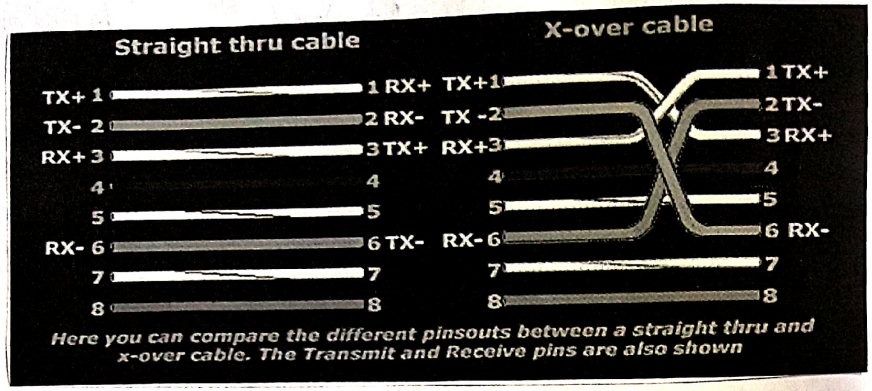
Category	data transmission	Advantage / Disadvantage	Application / Use
Category 6/6a	10 Gbps	× shielded × faster than UTP	Gigabit ethernet, 10G ethernet, widely used in data centre

3. SSTP

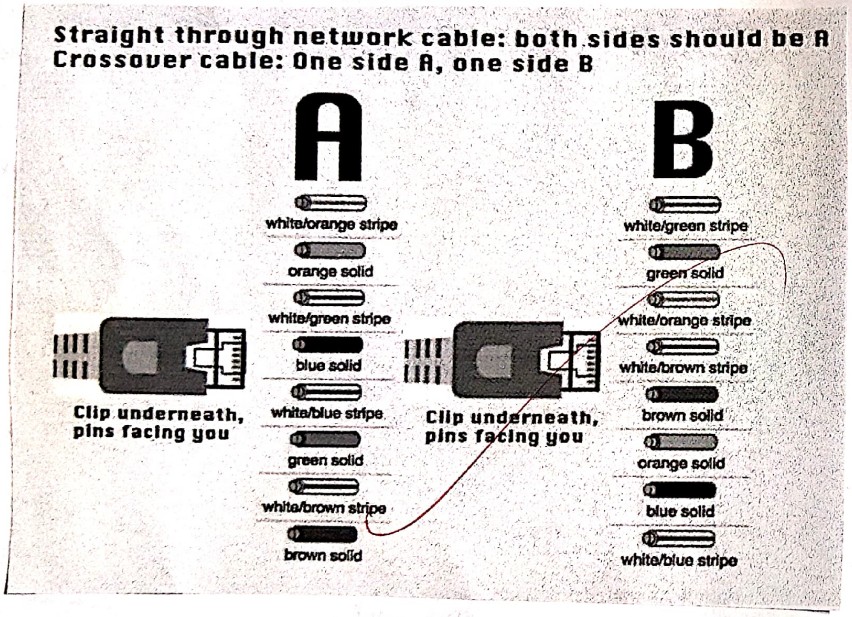
Category 7	10Gbps	× less susceptible to noise and interference Disadvantage: × expensive × greater installation effort	Gigabit ethernet, 10G ethernet, (100m)

4. Coaxial Cable

RG-6 RG-59 RG-11	10-100 Mbps.	× high bandwidth. × immune to interference × low loss bandwidth × versatile. Disadvantage: × limited distance × cost × size is bulky.	Speed of signal is 500 m. Television network, high speed internet connection



Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown



5. Fibre Optics Cable.

Single mode multi-mode	100 Gbps	* High speed, * high bandwidth, * high security, * long distance. Disadvantages: * expensive * requires skilled installers	* maximum distance of fibre optics cable is around 100 m

Make your Own Ethernet Cross-Over Cable.

Tools & Parts needed:

* ethernet cabling. Cat 5e is certified for gigabit support.
* A crimping tool. a all in one networking tool. shaped to push down the pins in the plug.
* 2 RJ45.
* optional 2 plug shields.

Step 1: To start construction of the device, begin by threading shields onto the cable.

Step 2: Next, strip approximately 15cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3: After, there should be 4 "twisted pairs"
One should be in arrangement A and other in B

Step 4: Once order is correct, bunch them together in a line..
The difficult aspect is placing these into the RJ45 plug without messing up the order.
clip side facing away from you and have gold pins facing toward you.

Step 5: Next, push the cable in.
The notch of the cable shielding and if it isn't, that means that you stripped off too much.

Step 6: insert into the crimping tool and push down.
It should be shaped correctly.

Step 7: Lastly, repeat for the other end using diagram B.
To test it, plug it in & attempt to connect two devices directly.

RESULT:
Thus, different types of network cables are studied.